

SURYANSH SINGH

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Education

Bhagwan Parshuram Institute of Technology (BPIT), Delhi

Delhi

B.Tech Computer Science Engineering

Aug 2024 - May 2028

– CGPA: 9.21/10

Woodland Academy

Gorakhpur

CBSE Class XII

2024

– Percentage: 92%

Projects

Genora – Sketch-to-Image — Next.js, TypeScript, Tldraw, Zustand — GitHub

Jan 2026

- Multi-step AI pipeline: sketch canvas export → vision analysis → prompt generation → image synthesis
- Client-side image compression: 1.1MB reduced to 31KB (97% reduction), enabling 32KB base64 limit of Puter.js vision endpoint
- State management: Zustand tracks editor instance, async processing status, and generated outputs across multi-step workflow
- Deployed with preview modal, style presets (Realistic, Anime, Cinematic), and download functionality

Synapz – AI Knowledge Platform — Next.js, Express.js, MongoDB, TypeScript — GitHub Live

Dec 2025

- Full-stack platform: users save notes and retrieve them via semantic search and RAG-based AI chat
- Semantic search: Cohere embeddings stored in MongoDB; query uses cosine similarity to find top-K relevant documents before LLM
- RAG pipeline integration: retrieved documents passed to Groq LLaMA with token management to prevent context overflow
- Backend: Express.js REST APIs, JWT authentication, MongoDB schema for notes and embeddings
- Frontend: React dashboard with Zustand state, real-time search results, and citation tracking

Clipper – Video Processing Tool — Go, Next.js, Docker, TypeScript — GitHub Live

Nov 2025

- Full-stack video processor: integrated yt-dlp for media acquisition from 1000+ platforms (YouTube, Instagram, TikTok)
- Go backend: subprocess management for FFmpeg video operations, concurrent request handling, Docker containerization
- AI pipeline: AssemblyAI transcription (50+ languages), Groq LLaMA summarization, automatic cleanup after 10 minutes
- Frontend: Next.js on Vercel, backend on Railway; handles video format conversion

Technical Skills

Languages: JavaScript, TypeScript, Go, C++, Python

Frontend: React, Next.js, Tailwind CSS, Zustand, Tldraw

Backend: Node.js, Express.js, Go, REST APIs

Databases & DevOps: MongoDB, PostgreSQL, Docker

AI/APIs: Cohere embeddings, Groq LLaMA, OpenRouter, AssemblyAI, Pollinations.ai

Deployment: Vercel, Railway

Achievements

ET-AI Hackathon 2026: Semi-Finalist for innovative AI solution design

Reverse Engineering Workshop: 3rd Position at Delhi Technological University